PHYS1408 Practice Exam 2-4 — Answer Key

Worked solutions

All answers were checked numerically with `_scripts/build_exam_2_4.py`. $g = 9.8$ m/s^2 throughout.

Practice Exam 2-4 uses the same problem types as Exams 2-1 through 2-3 with new numbers and new twists (find μ instead of a height, walk **out** instead of in, power from Fv, a one-mass ring pulley). Question 2 is a **minimum** trap: round up to the choice that works. The last column says which 3×5 notecard entry solves each question.

Q	Answer	Value	Topic	Notecard entry
1	**c**	15.5 m/s	Banked curve, maximum speed	Curve
2	**d**	18.75 m → 20	Loop-the-loop, minimum launch height	Loop
3	**a**	0.15	Halfpipe, find μ	Halfpipe
4	**e**	5.07 m/s	Sliding down a rough incline	One block on incline (Down)
5	**d**	0.20 m	Spring stops a sliding mass	Spring
6	**c**	2940 W	Power, $P = Fv$	Power
7	**b**	1.81 m/s	Incline + hanging mass + disk pulley	Incline + hanging mass + pulley
8	**c**	18.75 m/s	Completely inelastic collision	Stick (inelastic)
9	**e**	+4.8 m/s	1-D elastic collision	Elastic
10	**d**	193 m/s	Ballistic pendulum, bullet speed	Ballistic
11	**e**	90 rad	Rotational kinematics	Rot. kinematics
12	**b**	3.92 m/s^2	Hanging mass on a ring pulley	Atwood / mass on pulley (one mass)
13	**b**	3.27 m/s^2	Rolling acceleration	Rolling (disk row)
14	**a**	+1.23 rad/s	Counter-rotating object dropped on a disk	Angular momentum
15	**a**	1.22 rad/s	Person walks from the center to the rim	Angular momentum (walk in/out)

1. Banked curve, maximum speed — c) 15.5 m/s

Source: Ch 6 notes p. 3/7–5/7. **Notecard:** Curve.

At v_{max} the car tends to slide **up** the bank, so friction points **down** the bank. The notecard formula with the + signs:

$$v_{max} = \sqrt{gr\,\frac{\tan\theta + \mu_s}{1 - \mu_s \tan\theta}}$$

Plug in ($\tan 12^\circ = 0.2126$, $g = 9.8$ m/s^2, $r = 45$ m):

$$v_{max} = \sqrt{(9.8)(45)\,\frac{0.2126 + 0.30}{1 - (0.30)(0.2126)}}$$

$$= \sqrt{441 \times \frac{0.5126}{0.9362}} = \sqrt{241.4} = 15.5 \text{ m/s}$$

Distractors:

- a) 9.7 is the frictionless bank, $\sqrt{gr\tan\theta}$.
- b) 11.5 is the flat-curve formula, $\sqrt{\mu_s gr}$.
- d) 21.9 multiplies by an extra $\sqrt{2}$.
- e) 241 forgets the square root.

2. Loop-the-loop, minimum launch height — d) 20 m

Source: Ch 7/8 notes p. 12/18–13/18. **Notecard:** Loop.

Top of the loop: "barely" means $N = 0$, so $mg = \dfrac{mv_t^2}{R}$ and $v_t^2 = gR$.

Energy, start to top ($E_i = E_f$; the top is $2R$ high):

$$E_i = mgh$$

$$E_f = mg(2R) + \tfrac{1}{2}mv_t^2$$

$$mgh = 2mgR + \tfrac{1}{2}m(gR)$$

$$h = \tfrac{5}{2}R$$

Plug in:

$$h = \tfrac{5}{2}(7.5) = 18.75 \text{ m}$$

Answer-choice trap. 18.75 m is a **minimum**. Starting at 18 m, the car falls off the track, even though 18 is the closest number. The smallest choice that works is **d) 20 m**.

Distractors:

- a) 11 m is $\frac{3}{2}R$.
- b) 15 m is $2R$: it reaches the top with zero speed and falls.
- c) 18 m is the nearest number, but it's below the minimum.
- e) 38 m is $5R$ (mixing up $v_b^2 = 5gR$ with a height).

3. Halfpipe, find μ — a) 0.15

Source: Ch 7/8 notes p. 4/18–5/18. **Notecard:** Halfpipe.

Notecard (from rest, one crossing): $h_f = h_i - \mu L$. **Solve for μ:**

$$\mu L = h_i - h_f$$

$$\mu = \frac{h_i - h_f}{L}$$

Plug in:

$$\mu = \frac{4.0 - 3.1}{6.0} = \frac{0.9}{6.0} = 0.15$$

Distractors:

- b) 0.52 is h_f/L.
- c) 0.67 is h_i/L.
- d) 0.78 is h_f/h_i.
- e) 0.90 is the height lost, not divided by L.

4. Sliding down a rough incline — e) 5.07 m/s

Source: Ch 7/8 notes p. 10/18. **Notecard:** One block on incline (Down).

Notecard, sliding **down**:

$$v_f = \sqrt{v_i^2 + 2g\,\Delta x(\sin\theta - \mu\cos\theta)}$$

Plug in ($\sin 35^\circ = 0.5736$, $\cos 35^\circ = 0.8192$, $\Delta x = 3.0$ m along the slope):

$$v_f = \sqrt{2.0^2 + 2(9.8)(3.0)\,(0.5736 - 0.25 \times 0.8192)}$$

$$= \sqrt{4.0 + 58.8(0.3688)} = \sqrt{4.0 + 21.68} = \sqrt{25.68} = 5.07 \text{ m/s}$$

The bracket is positive, so the block speeds up.

Distractors:

- a) 2.0 m/s assumes the speed doesn't change.
- b) 3.85 forgets the 2 in $2g$.
- c) 4.66 forgets v_i^2.
- d) 4.80 uses friction μmg instead of $\mu mg\cos\theta$.

5. Spring stops a sliding mass — d) 0.20 m

Source: Ch 7/8 notes p. 11/18. **Notecard:** Spring.

Level floor, so $h = 0$. The notecard's "stop a mass" line becomes

$$x = \sqrt{\frac{2mgh + mv_i^2}{k}} = \sqrt{\frac{mv_i^2}{k}}$$

(That's $\frac{1}{2}kx^2 = \frac{1}{2}mv_i^2$: all the KE becomes spring PE.)

Plug in:

$$x = \sqrt{\frac{(1.5)(4.0)^2}{600}} = \sqrt{\frac{24}{600}} = \sqrt{0.040} = 0.20 \text{ m}$$

Distractors:

- a) 0.010 is mv/k (momentum, not energy).
- b) 0.040 forgets the square root.
- c) 0.14 drops a $\frac{1}{2}$ on one side only.
- e) 0.28 has an extra factor of 2.

6. Power, $P = Fv$ — c) 2940 W

Source: Ch 7/8 notes p. 14/18. **Notecard:** Power.

At constant speed the motor's force equals the weight, $F = mg$. Notecard: $P = Fv$.

$$P = Fv = mgv$$

Plug in:

$$P = (250)(9.8)(1.2) = 2940 \text{ W}$$

Distractors:

- a) 180 W is $\frac{1}{2}mv^2$ (energy, not power).
- b) 300 is mv (momentum).
- d) 5880 doubles it.
- e) 29,400 uses 12 m/s.

7. Incline + hanging mass + disk pulley — b) 1.81 m/s

Source: Ch 7/8 notes p. 15/18; 2015 Exam 2 #13. **Notecard:** Incline + hanging mass + pulley.

Notecard, with m_2 falling (m_1 slides **up** the slope):

$$v_f = \sqrt{v_i^2 + \frac{2g\,\Delta x\,B}{m_1 + m_2 + I/R^2}}$$

$$B = m_2 - m_1 \sin\theta - \mu m_1 \cos\theta$$

Net pull ($\sin 20^\circ = 0.3420$, $\cos 20^\circ = 0.9397$):

$$B = 5.0 - 6.0(0.3420) - 0.20(6.0)(0.9397) = 5.0 - 2.052 - 1.128 = 1.820 \text{ kg}$$

Pulley term: a disk has $I/R^2 = \frac{1}{2}M = 2.0$ kg.

Plug in ($v_i = 0$, $\Delta x = 1.2$ m):

$$v_f = \sqrt{0 + \frac{2(9.8)(1.2)(1.820)}{6.0 + 5.0 + 2.0}}$$

$$= \sqrt{\frac{42.81}{13.0}} = \sqrt{3.293} = 1.81 \text{ m/s}$$

Distractors:

- a) 1.69 treats the pulley as a ring ($I/R^2 = M$).
- c) 1.97 ignores the pulley's mass.
- d) 2.31 ignores friction.
- e) 4.85 is free fall, $\sqrt{2g\Delta x}$.

8. Completely inelastic collision — c) 18.8 m/s

Source: Ch 9 notes p. 1/5. **Notecard:** Stick (inelastic).

They stick, so momentum only. Notecard:

$$v_f = \frac{m_1 v_1 + m_2 v_2}{m_1 + m_2}$$

Plug in (both moving the same way, so both velocities are +):

$$v_f = \frac{(1500)(25) + (2500)(15)}{1500 + 2500} = \frac{37{,}500 + 37{,}500}{4000} = 18.75 \text{ m/s}$$

Distractors:

- a) 0 treats the truck as moving the opposite way.

- b) 9.4 leaves out the truck's momentum.
- d) 20.0 averages the two speeds.
- e) 40.0 adds them.

9. 1-D elastic collision — e) +4.8 m/s

Source: Ch 9 notes p. 2/5–3/5. **Notecard:** Elastic.

Notecard formulas, right = +: $m_1 = 4.0$, $v_{1i} = +3.0$; $m_2 = 1.0$, $v_{2i} = 0$.

$$v_{1f} = \frac{(m_1 - m_2)v_{1i} + 2m_2 v_{2i}}{m_1 + m_2} = \frac{(3.0)(3.0) + 0}{5.0} = +1.8 \text{ m/s}$$

$$v_{2f} = v_{1i} + v_{1f} - v_{2i} = 3.0 + 1.8 - 0 = +4.8 \text{ m/s}$$

A heavy ball hitting a light one sends the light one off faster than the heavy one was going.

Distractors:

- a) 0 confuses it with the stuck ball case.
- b) +1.8 is the 4.0 kg ball's velocity.
- c) +2.4 is the stick-together answer.
- d) +3.0 is the equal-mass (cue ball) result.

10. Ballistic pendulum, bullet speed — d) 193 m/s

Source: Ch 9 notes p. 5/5. **Notecard:** Ballistic.

Notecard: $v = \dfrac{M + m}{m}\sqrt{2gh}$ (momentum through the hit, energy for the swing).

Plug in:

$$v = \frac{1.5 + 0.012}{0.012}\sqrt{2(9.8)(0.12)}$$

$$= 126 \times 1.534 = 193 \text{ m/s}$$

Distractors:

- a) 1.5 m/s is $V = \sqrt{2gh}$, the block's speed after the hit.
- b) 17 m/s uses energy conservation through the collision (never do that).
- c) 126 is just the mass ratio $\frac{M+m}{m}$.
- e) 386 doubles it.

11. Rotational kinematics — e) 90 rad

Source: Ch 10 notes p. 1/8; Ch 11 notes p. 6/8. **Notecard:** Rot. kinematics.

α isn't given or asked for, so use the equation without it:

$$\Delta\theta = \frac{\omega_i + \omega_f}{2}\, t$$

Plug in:

$$\Delta\theta = \frac{30 + 0}{2}(6.0) = 15 \times 6.0 = 90 \text{ rad}$$

Distractors:

- a) 5.0 is the angular acceleration (rad/s^2).
- b) 14.3 is the answer in revolutions, $90/2\pi$.
- c) 15 is the average angular velocity.
- d) 45 halves it twice.

12. Hanging mass on a ring pulley — b) 3.92 m/s^2

Source: Ch 10 notes p. 4/8; 2015 Exam 2 #9. **Notecard:** Atwood / mass on pulley (one mass).

Notecard, one mass: $a = \dfrac{mg}{m + I/R^2}$. A ring has $I = MR^2$, so $I/R^2 = M$ (R cancels).

$$a = \frac{mg}{m + M}$$

Plug in:

$$a = \frac{(2.0)(9.8)}{2.0 + 3.0} = \frac{19.6}{5.0} = 3.92 \text{ m/s}^2$$

Distractors:

- a) 2.45 uses $I/R^2 = 2M$.
- c) 5.60 treats the pulley as a disk ($\frac{1}{2}M$).
- d) 6.53 leaves m out of the denominator.
- e) 9.80 is free fall (massless pulley, nothing holding it).

13. Rolling acceleration — b) 3.27 m/s²

Source: Ch 10 notes p. 3/8. **Notecard:** Rolling (disk row).

Notecard, disk (solid cylinder) row: $a = \frac{2}{3} g \sin\theta$.

Plug in:

$$a = \tfrac{2}{3}(9.8)\sin 30^\circ = \tfrac{2}{3}(9.8)(0.5) = 3.27 \text{ m/s}^2$$

Distractors:

- a) 2.45 is a ring, $\frac{1}{2} g \sin\theta$.
- c) 3.50 is a sphere, $\frac{5}{7} g \sin\theta$.
- d) 4.90 is sliding, $g \sin\theta$.
- e) 6.53 forgets $\sin\theta$.

14. Counter-rotating object dropped on a disk — a) +1.23 rad/s

Source: Ch 11 notes p. 3/8; 2015 Exam 2 #11. **Notecard:** Angular momentum.

$L_i = L_f$, with the ring's ω_2 **negative**. Notecard:

$$\omega_f = \frac{I_1\omega_1 + I_2\omega_2}{I_1 + I_2}$$

Moments of inertia first:

$$I_1 = \tfrac{1}{2}(150)(2.0)^2 = 300 \text{ kg} \cdot \text{m}^2$$

$$I_2 = (40)(1.5)^2 = 90 \text{ kg} \cdot \text{m}^2$$

Plug in:

$$\omega_f = \frac{(300)(2.5) + (90)(-3.0)}{300 + 90} = \frac{750 - 270}{390} = +1.23 \text{ rad/s}$$

Distractors:

- b) +1.78 treats the ring as a disk ($\frac{1}{2}mr^2$).
- c) +1.92 ignores the ring's spin.
- d) +2.50 is no change.
- e) +2.62 forgets the minus sign on the ring.

15. Person walks from the center to the rim — a) 1.22 rad/s

Source: Ch 11 notes p. 7/8. **Notecard:** Angular momentum (walk in/out).

Notecard, walk in/out: $\omega_f = \dfrac{I_i}{I_f}\omega_i$. At the center the person adds nothing ($m \cdot 0^2$); at the rim they add mR^2.

$$\omega_f = \frac{\frac{1}{2}MR^2}{\frac{1}{2}MR^2 + mR^2}\,\omega_i = \frac{M}{M+2m}\,\omega_i$$

(That's the notecard's rim→center ratio, flipped, because they're walking **out**.)

Plug in:

$$\omega_f = \frac{250}{250+2(60)}(1.8) = \frac{250}{370}(1.8) = 1.22 \text{ rad/s}$$

Walking outward adds I, so the merry-go-round slows down.

Distractors:

- b) 1.45 forgets the $\frac{1}{2}$ on the disk.
- c) 1.80 is no change.
- d) 2.23 uses $\frac{M+m}{M}$, a speed-up.
- e) 2.66 uses the rim→center formula without flipping it (walking in, not out).